DTL–ST–REPRO–001

Independent Reconstruction of Noise-Driven Discrete Time Liquids

Pre-Execution Protocol Candidate v0.1.0

Original spin–cavity reference first; typed Spiral-Time extensions only in later, separate protocols

NOT FROZEN | **NO MANY-BODY RUN EXECUTED** | **NO PHYSICAL VALIDATION**

Proposed author	Marcel Krüger; Independent Researcher, Germany. Protocol designer; independent external review pending.
Date / identity	2026-11-02; DTL-ST-REPRO-001 candidate v0.1.0, separate from Mathematical Core v0.3.1.
Source of physics	Wang, Schindler, Bukov and Michael, *Discrete time liquids*, arXiv:2610.30266v1, 18-page manuscript including supplement (21 October 2026), hereafter [W26].
Data provenance	Original simulation/code Zenodo DOI 10.5281/zenodo.23168407, currently restricted. Original manuscript PDF locally available.
Decision authority	No change to physical equations, samples, success thresholds, or null families after confirmatory target exposure. Freeze requires scientific review, data/parameter manifest, runnable implementation and costs.

Executive scientific decision

This document specifies what it would take to *independently reconstruct* the published classical spin–cavity dynamics and assess their characteristic period doubling and time-soliton statistics. The published paper makes the source model sufficiently explicit for an independent implementation; unavailable original input files/code, unrecovered figure-point arrays and random streams still prevent an identical archived-code replay. The outcome of Manvitha B.'s separate kinetic-oscillator audit is *not* evidence for a many-body time liquid. Its original energy-ledger FAIL and subsequent diagnosis remain separate provenance objects.

Scope barrier: No HLV/Spiral-Time phase-memory feature or kinetic coefficient may be added in DTL-ST-REPRO-001. A later diagnostic study and later kinetic-intervention study each require a separate preregistered freeze. The square spin lattice is not the HLV 6D cut-and-project carrier; A_Θ is a coefficient, not a fifth operator.[K26]

1 Source-native model and exact reconstruction target

For a square lattice of $L \times L$ classical unit spins with periodic spatial boundaries, $\mathbf{S}_i \in S^2$, the main-text Hamiltonian and modulation are [W26, Eqs. (3)–(4)]:

$$H_{\mathrm{spin}}(t) = -J_{\mathrm{eff}}(t) \sum_{\langle ij \rangle} S_i^z S_j^z - g_{\mathrm{eff}}(t) \sum_i S_i^x, \tag{1}$$

$$J_{\mathrm{eff}}(t) = J[1 + \lambda y(t)], \qquad g_{\mathrm{eff}}(t) = g_0[1 - \lambda y(t)], \qquad g_0 = \frac{\pi + \varepsilon_0}{\tau_{\mathrm{osc}}}. \tag{2}$$

The spin equation [W26, Methods Eq. (12)] is

$$\dot{\mathbf{S}}_i = \mathbf{S}_i \times \mathbf{B}_i(t), \quad \mathbf{B}_i = \left(g_{\mathrm{eff}}, 0, J_{\mathrm{eff}} \sum_{j \in \mathrm{nn}(i)} S_j^z \right). \tag{3}$$

The oscillator is *independent of the spins in the main reference model* [W26, Eq. (5)]:

$$\ddot{y} + \gamma\dot{y} + \omega_0^2 y = \xi(t), \quad \mathbb{E}[\xi(t)\xi(s)] = D\delta(t-s), \quad D = \frac{2\gamma k_B T_{\text{bath}}}{M_{\text{osc}} L^2}, \tag{4}$$

$$\omega_c = \sqrt{\omega_0^2 - (\gamma/2)^2} = \frac{2\pi}{\tau_{\text{osc}}}, \quad \tau_{\text{cav}} = \frac{2}{\gamma}, \quad T_0 = \frac{M_{\text{osc}}\omega_0^2 L^2}{k_B}. \tag{5}$$

The source sets $J = 5$, $\lambda = M_{\text{osc}} = k_B = 1$, $L = 100$, $\theta_{\text{init}} = 0.2$ for the principal figures. The damping-dependent ω_0 is recalculated to hold the damped oscillation period fixed; it is *not* held at the same value when γ changes. T_{bath}/T_0 is a dimensionless temperature control, which must be mapped back to D using the stated T_0. A missing factor of L^2, mass or ω_0^2 is a substantive physics failure. For $\gamma > 0$, the stationary source oscillator implies the analytic control identities $\mathbb{E}[y^2] = D/(2\gamma\omega_0^2) = T_{\text{bath}}/T_0$, $\mathbb{E}[\dot{y}^2] = D/(2\gamma)$, and $\mathbb{E}[y\dot{y}] = 0$. Their stochastic uncertainty, finite warm-up and autocorrelation require explicit treatment.

2 Initial conditions, discretization and stochastic semantics

Spins. Sample independently $\phi_i \sim U(0, 2\pi)$ and $\cos\theta_i \sim U(\cos\theta_{\text{init}}, 1)$, set the unit spin as in [W26, Eq. (14)]:

$$\mathbf{S}_i(0) = (\sin\theta_i \cos\phi_i,\ \sin\theta_i \sin\phi_i,\ \cos\theta_i). \tag{6}$$

The source uses an exact-rotation symmetric Strang split: x rotation through a half transverse-field step, z rotation through the full neighbor-interaction step with S_j^z fixed in that substep, and another x half rotation. Rotation signs must be checked against $\dot{\mathbf{S}} = \mathbf{S} \times \mathbf{B}$ rather than assumed from a graphics rotation API. Spin norm and boundary-neighbor census must be checked every numerical stage.

Cavity. Start $y = \dot{y} = 0$ and pre-evolve for $5/\gamma$ before recording, for $\gamma > 0$. The source uses, during each time step Δt, an independent standard normal η_n and the piecewise-constant random force $\xi_n = \sqrt{D/\Delta t}\,\eta_n$, so that its *time-integrated* impulse has variance $D\Delta t$. Any replacement by an exact Ornstein–Uhlenbeck update or alternative stochastic solver must be recorded as an *independent algorithmic comparison*, not silently identified with the source integrator.

Source resolutions. The Methods report at least 20 steps per period: for $\tau_{\text{osc}} = 0.2$, $\Delta t = 0.005$ (40 steps); for 0.01, $\Delta t = 0.0005$ (20 steps); for 0.001, $\Delta t = 0.00005$ (20 steps). The main figures use $N_p = 10^4$ periods and $N_{\text{traj}} = 100$ independent noise realizations unless otherwise specified. [W26, Methods and Supplement S11]. These are *source values*, not currently executable performance claims.

3 Reference observables and source-figure checkpoints

Direct measurements. Store per-trajectory stroboscopic $M^z(n) = L^{-2}\sum_i S_i^z(n\tau_{\text{osc}})$, the staggered signal $(-1)^n M^z(n)$, oscillator y and $\dot{y}$ summaries, random seed identity, time-grid identity and quality masks. Preserve a source-faithful definition of the normalized $s(n)$ [W26, Eq. (6)] and implement the correlation and thermalization times [W26, Eqs. (7)–(11)] from source-aligned definitions, not ad hoc change-point outputs. The original independent-flip toy model is

$$\sigma_n = \sigma_{n-1} b_n, \quad \Pr(b_n = -1) = p, \quad s_n = e^{-n\tau_{\text{osc}}/\tau_{\text{th}}}\sigma_n. \tag{7}$$

The Bernoulli approximation *belongs to the published null model*; its predictive independence must not be inferred for observed many-body solitons.

Primary source-figure target A (Fig. 2). At $\tau_{\text{osc}} = 0.2$, $\gamma = 10^{-3}$, $\varepsilon_0 = 0.1$, contrast $T_{\text{bath}}/T_0 =$ 0.01 and 0.9 using identical operational definitions. The source reports a weak/no subharmonic signal at low temperature and sustained period doubling in the 0.9 case. Quantify the $\nu = \pi$ component

relative to predeclared neighboring spectral bands, with Monte Carlo uncertainty. Source Figure 2 is a *qualitative target until original numerical arrays become available.*

Primary target B (Fig. 3). At $\tau_{\text{osc}} = \varepsilon_0 = 0.001$, $T_{\text{bath}}/T_0 = 1$, compare $\tau_{\text{cav}}/\tau_{\text{osc}} = 20$ vs 200 and test the source signature: recurrent phase reversals with local period doubling, and separated correlation/thermalization times under the shorter coherence time. Source Figure 3 uses $L = 100$, $N_p = 10^4$, $N_{\text{traj}} = 100$. Avoid replacing a visibly fluctuating magnetization with a toy Bernoulli series and labeling it a spin reproduction.

Secondary target C (Fig. 1b). At $\tau_{\text{osc}} = 0.2$, $\gamma = 10^{-4}$, $\varepsilon_0 = 0.1$, reproduce the temperature-dependent ensemble Fourier amplitude $|\widetilde{F}[\mathbb{E}(M^z)](\nu)|$ and its isolated $\nu = \pi$ subharmonic response. Use the same ensemble-first convention.

Secondary target D (Fig. 1c and Fig. 4). Reconstruct the source phase-classification logic and the distribution of individual-trajectory correlation times. The original numerical region map and histogram bins are not supplied in the paper, hence exact pixel/array agreement is `BLOCKED` until inputs/arrays are obtained. The source classification uses $\langle\langle\tau_{\text{corr}}\rangle\rangle/\tau_{\text{th}} > 0.418$ for its DTC/DTL split, and $\langle\langle\tau_{\text{corr}}\rangle\rangle < 10\tau_{\text{osc}}$ for the disordered-region cutoff; these are *source empirical conventions, not universal physical thresholds.* Its supplement reports sensitivity of disordered counts to changing the latter threshold.

4 Independent validation ladder – no outcome-driven changes

1. **F0: Provenance/identity.** Hash [W26] PDF; freeze extracted Eqs. (3)–(5), (12)–(16), essential equations (6)–(11), and Supplement S11; record source commit, repository license, exact code inputs, acquisition/permissions and all deviations. If source code or data remain restricted, mark archived-code replay `BLOCKED`; allow independent reconstruction under that exact name.
2. **F1: Single-stage oscillator unit tests.** Verify ω_c, τ_{cav}, temperature/noise variance and stationary covariance from S2 by direct Monte Carlo; compare sampled impulse variance $D\Delta t$ with theoretical confidence intervals and inspect pre-equilibration effects. The source-only Langevin dynamics must be verified *before* spins are switched on.
3. **F2: Deterministic spin mechanics.** Compare Strang integration against an independently implemented adaptive solver on a predefined smooth deterministic drive. Check periodic boundaries, neighbor sums, exact spin-norm conservation within a numerical error budget, $\mathcal{O}(\Delta t^2)$ convergence where an asymptotic regime is resolved, and source Supplement Fig. S10's numerical behavior. No random forcing in this gate.
4. **F3: Stochastic convergence.** On identical Brownian paths across resolutions (coarse increments sum fine increments), assess weak-observable stabilization, uncertainties and comparison with a separately implemented stochastic integrator. Do not expect pathwise equality for simulations driven by *independent* noise sequences at different step sizes.
5. **F4: Feasibility commitment.** Before production: timed kernels, peak RAM, serialized outputs, independent-trajectory chunking, exact resume/no-overwrite behavior, deterministic seed-to-trial mapping and environment lock. The locked run cannot be replaced by a fast demonstrator. Any departure from $L = 100$, $N_p = 10^4$, $N_{\text{traj}} = 100$ creates a different finite-size/sampling study.
6. **F5: Original-physics reproduction.** Run primary figure targets A and B with full declared conditions; record all trajectories, aggregate uncertainty and numerical quality; perform secondary targets as resources permit. The paper's main qualitative phenomena must not be inferred from a single selected favorable trajectory.
7. **F6: Independent replication.** A second implementation or lab compares source-native observables at matched physical/numerical conditions, with random-seed-independent statistical comparisons. A reproduced figure-like pattern alone does not validate equilibrium physics or the original experimental existence (no experiment exists in this study).

5 Pre-specified operational gates and uncertainty

Gate	Expected outcome and decision rule (candidate; quantitative statistical cutoffs not yet frozen)
G00 Source provenance	Source equations, parameters, temperature/noise units, figure conditions and hashes mapped exactly. Mismatch: FAIL. Restricted archived code: BLOCKED for strict replay.
G01 Spin kinematics	Periodic neighbor and spin sign tests correct; normalized exact rotations preserve $\lvert\mathbf{S}_i\rvert$ with engineering drift at most 10^{-10}. Technical FAIL if not.
G02 Deterministic integration	Strang second-order convergence over at least three resolved step sizes, independently checked against adaptive ODE integration; FAIL/INCONCLUSIVE if numerical floor or unresolved sensitivity.
G03 Stochastic cavity	Stationary variance, impulse variance and autocorrelation agree with analytic laws within dependence-aware Monte Carlo intervals. Numerical FAIL if a well-resolved mismatch persists.
G04 Source Fig. 2	Subharmonic spectral statistic with uncertainty for both $T/T_0 = 0.01$ and 0.9 at source settings. PASS requires a well-resolved increased subharmonic response at intermediate temperature versus low temperature, subject to a quantitative gate fixed before production; otherwise FAIL or INCONCLUSIVE.
G05 Source Fig. 3	Verify sign-switch evidence on full $s(n)$ trajectories plus reported ensemble correlation, mean single-trajectory correlation, and thermalization scales at cavity ratios 20 and 200; quantify uncertainty. Without a locked quantitative gate, report descriptive reconstruction, not confirmatory PASS.
G06 Sample/resource	Source-scale ($N_p = 10^4$, $N_{\mathrm{traj}} = 100$, $L = 100$) with unfiltered realizations and convergence reporting. If not feasible, BLOCKED; reduced scale cannot inherit original-reproduction PASS.
G07 Reproducibility	Complete parameter, seed, software and hardware manifest, immutable logs, outputs and SHA-256 verification. Missing provenance: FAIL for replayability.
G08 Claim boundary	Do not infer quantum-spin, equilibrium, fundamental-time or HLV-specific physical claims from a simulated metastable classical model. Unsupported promotion is a claim-boundary FAIL.

Threshold provenance. 10^{-10} in G01 is an *engineering numerical tolerance*, not a physical constant; its adequacy must be verified against machine precision and run length. The published 0.418 and $10\tau_{\mathrm{osc}}$ are source-defined classification conventions, not our newly optimized thresholds. The 95% uncertainty interval and $N_{\mathrm{traj}} = 100$ are sampling standards/source sample size, not proof that 100 runs are statistically sufficient for every rare-soliton tail. A result meeting selected gates while failing any load-bearing physical reproduction is not an overall PASS.

Uncertainty. Independence unit is the full stochastic trajectory, never adjacent periods from the same realization. Provide trajectory-level bootstrap with preregistered seed and ≥ 2000 resamples; use simultaneous adjustment for explicitly multiple primary spectral or timescale comparisons, or report them as exploratory until a familywise rule is frozen. For rare events with few or zero solitons, give exact or interval-censored rate limits; *no* spurious precise estimates. Primary qualitative pattern matching should additionally be supported by convergence and sensitivity.

Result labels. `PASS`: all frozen eligible load-bearing gates met. `FAIL`: a valid run decisively violates any required claim/gate. `INCONCLUSIVE`: correct execution but insufficient power, unstable numerics, or unresolvable source ambiguity. `BLOCKED`: missing legal source data, resources, critical parameters, or code. `INVALID`: run breaks the frozen implementation/quality contract; no scientific verdict. A pre-freeze technical feasibility exercise never counts as the frozen confirmatory run.

6 Compute cost, resource controls and original-code access

For $L = 100$, $N_{\mathrm{traj}} = 100$, $N_p = 10^4$ and 20–40 substeps per period, the production demand is roughly

$$L^2 \times N_{\mathrm{traj}} \times N_p \times N_{\mathrm{step/period}} \approx 2 \times 10^{11} \text{ to } 4 \times 10^{11} \text{ spin-site time-step updates per parameter point,} \tag{8}$$

excluding cavity equilibration, parameter sweeps and multi-resolution comparisons. The product is an operation-count proxy, not runtime. A full 2D phase map may be substantially more expensive. Compute time may be outside a routine Colab session and is not claimed feasible without measured kernel timing and resumable, validated chunking. A CPU-only Python loop over lattice sites would be unsuitable; tested JIT/vectorized or GPU exact-rotation kernels may be needed.

The original Zenodo 23168407 is restricted; a request for permission is appropriate. Preserve original SHA-256 and terms of access if granted. Never regenerate "original" HEK-like data or treat new pseudorandom paths as historical author paths. Restricted original data do *not* block *independent reconstruction of the published equations*; they do block strict data/code identity claims.

7 Separation from subsequent Spiral-Time work

Phase	Allowed work and prohibited inference
R: this protocol	Original $A \equiv 1$ oscillator, original spins and original stochastic bath; no Spiral-Time modifications.
D: later ST-DIAG	Apply causal U_2 phase features and U_3 history features *only after* a reference reproduction, with no-leakage, iid-Bernoulli/no-memory, AR/state-space, HMM and matched finite-memory controls; predeclared out-of-sample forecast horizon and confidence intervals. A history feature cannot predict iid future flips beyond the base rate.
K: later ST-KIN	Posit new exogenous $A_\Theta(t) \in C^1$, $0 < a_- \le A \le a_+$ in $d(A\dot{y})/dt + \gamma\dot{y} + \omega_0^2 y = \xi_\Theta$. Specify bath convention and kinetic energy balance *first*. The divided equation is an ordinary time-varying oscillator with effective damping, stiffness and force; match these ordinary effects. Do not claim thermal equilibrium under unverified fluctuation–dissipation assumptions.
Native host	No use of HLV native Window/Golden Carrier without a mathematical and physical interface from $\mathbb{Z}^6$ to the $L \times L$ spin phase space. Such an interface is neither provided by Wang et al. nor established by v0.3.1.

8 Required pre-freeze sign-off and handover

Candidate review blockers: (1) extract a fully equation-accurate observable and source-temperature mapping including supplementary branch conventions; (2) demonstrate unbiased noise numerical equivalence, deterministic spin integrator correctness, and stochastic convergence; (3) resolve full-run compute capacity under source scale; (4) preserve the restricted-data access status and source-array limitations; (5) prepare an immutable source/parameter/seed manifest and exact figure targets; (6) secure review of gate thresholds and power by at least one independent reader.

At handover, produce `FROZEN_CONFIG.json`, `PROVENANCE.json`, `SHA256SUMS.txt`, exact source PDF hash, versioned read-only implementation/notebook, pre-run test logs, numerical budgets and sign-off date. **No primary scientific run is authorized by this candidate.** If the original source code arrives after a freeze, compare its identity separately; an altered original-code branch requires an explicitly named successor, not substitution inside the locked run.

References and provenance

[W26] Z. Wang, P. M. Schindler, M. Bukov, M. H. Michael, *Discrete time liquids*, arXiv:2610.30266v1 (21 October 2026); main Eqs. (3)–(16), Methods, Supplemental S1–S11, DOI 10.48550/arXiv.2610.30266; restricted data/code DOI 10.5281/zenodo.23168407.

[K26] M. Krüger, *Typed Spiral-Time Diagnostics and Positive Kinetic Modulation in Discrete Time-Liquid Models*, Mathematical Core v0.3.1, 2 November 2026, local author-review candidate; type-separation, conditional deterministic/stochastic energy and identifiability results. Related definition sources: *Spiral-Time Master Operator Family* v1.0.1 and *Cut-and-Project/Incidence-Spectral Core* v2.8.0.

Evidence boundary: All source-figure assertions above are explicit statements of [W26], not independent new observations. No original-code replay, production spin run, frozen confirmatory result, thermodynamic equilibrium confirmation or new HLV physics is claimed here.